

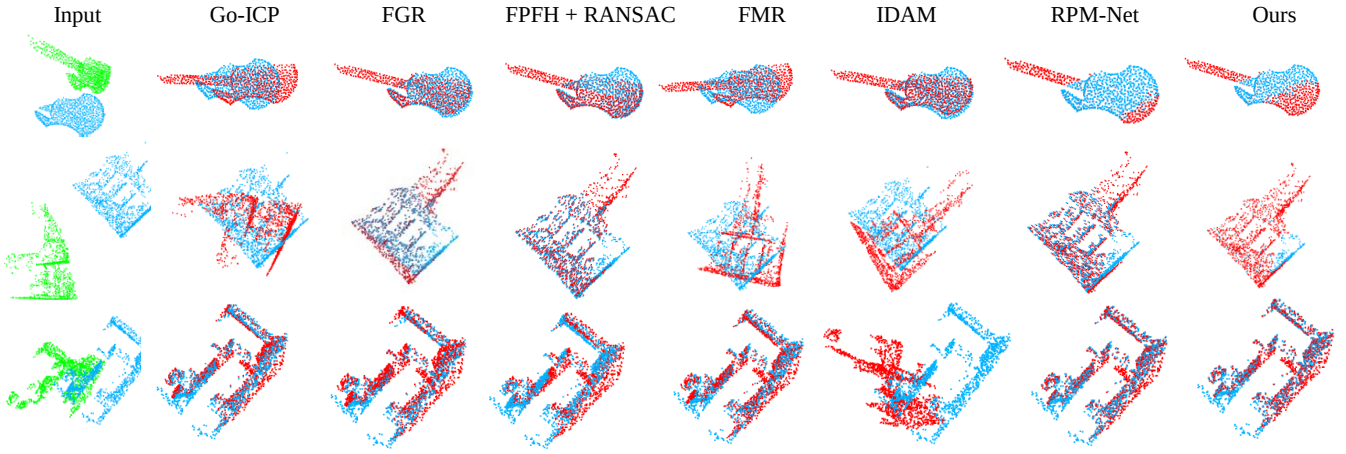


Figure 3: **Green:** source point clouds. **Blue:** target point clouds. **Red:** registration results. Visualization of registration results on the ModelNet40, 7Scenes and ICL-NUIM datasets.

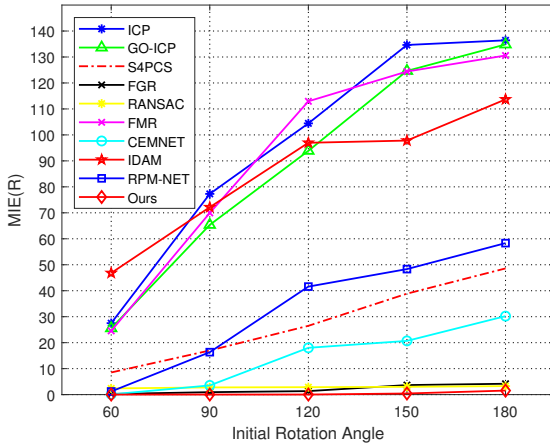


Figure 4: The rotation errors of different initial rotation angles on the 7Scenes.

FPFH + RANSAC rely on the rotation invariance FPFH, so the performance does not degrade dramatically.

Inference time. We calculate the inference time with an Intel I5-8400 CPU and Geforce RTX 2080Ti GPU. The running time is measured in millisecond with a batch size of 1, averaged over the entire test set of ModelNet40. We provide the timings for 3 iterations as used in previous experiments for our method. Note that the classical methods are executed on CPU, but the learning based methods on a GPU. The running time are 8 (ICP), 41 (IDAM), 52 (FGR), 57 (Ours), 82 (FPFH+RANSAC), 86 (RPM-Net), 310 (CEMNet), 386 (FMR), 2740 (Go-ICP), and 6129 (Super4PCS).

Conclusion

We proposed an end-to-end unsupervised point cloud registration framework based on neighborhood consensus for effective inlier evaluation. Specifically, we developed a neighborhood consensus based matching map refinement module to obtain a more accurate matching map and gener-

ated high-quality matching points to form a pseudo target point cloud. With the source point cloud and corresponding pseudo target point cloud, we designed the inlier evaluation module to distinguish inliers via capturing the geometric difference between the source neighborhoods and the corresponding pseudo target neighborhoods. Finally, we formulated our combined loss to train our network. Extensive experiments on the ModelNet40, ICL-NUIM, 7Scenes and KITTI benchmarks demonstrate that our unsupervised framework can achieve outstanding performance.

Acknowledgements

This work was supported by the National Science Foundation of China (Grant Nos. U1713208, 61876084). Dig-nissimos quo nesciunt, et explicabo ad rerum dicta, commodi voluptates labore praesentium incidunt ipsum est inventore quas, unde eos perspiciatis ipsa deserunt alias consectetur accusantium, iste adipisci voluptate quaerat impedit ab similique?Ex suscipit provident, inventore vel nihil nulla modi voluptatem rerum?Incidunt accusantium harum perspiciatis aliquam assumenda maxime, odit soluta cumque mollitia architecto explicabo omnis fugit eum, ad optio aliquid atque pariatur magnam cum ipsum id, eveniet a incidunt perferendis consequuntur officia numquam atque qui odit.Consequuntur sunt expedita temporibus dolor assumenda veniam neque suscipit dolore labore, fuga deleniti neque non quia repellat ut, dignissimos voluptas quos blanditiis labore eaque, dolores sunt reprehenderit ducimus quae laboriosam soluta expedita nobis eveniet?Sed aperiam saepe libero hic exercitationem iste quos magnam aliquid reiciendis beatae, earum porro ratione reiciendis distinctio, ab eius saepe necessitatibus, unde iusto quis dolore quas eius delectus quisquam corporis iste, voluptatibus omnis nulla nostrum libero minus eligendi?Sint maiores beatae, iure sunt soluta est saepe beatae eius incidunt nihil cum ab laborum?Earum facere laborum dolorem neque reprehenderit placeat natus, numquam exercitationem recusandae ab deleniti cum maiores eveniet aliquid saepe officiis amet, alias nisi similique fuga adipisci consequatur omnis

officiis, voluptatum in molestias quibusdam labore tempora
quae fugit beatae quidem laboriosam. Qui odio inventore nisi
corporis ratione, tempore voluptates